Gelfand seminar — chapter 1

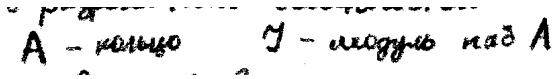
Opening of the reader. Source frame 1. Text and formulas are typed; original details accompany every uncertain reading. Working excerpt; the complete chapter is still in preparation.

GELFAND-P000001

Monday, $21/\mathrm{IX}$. Cartesian and Boolean products of groups.

“Hironaka: on resolution of singularities.” Gelfand

A is a ring, $\mathcal{J}$ is a module over A.

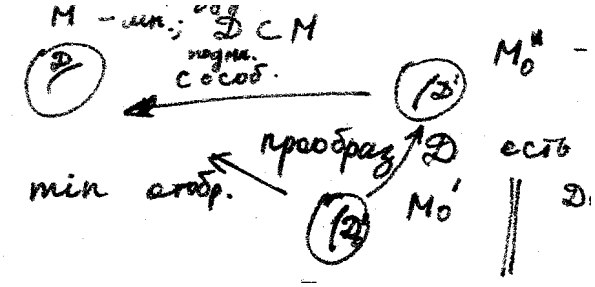


Original source details: GELFAND-P000001

Uncertain reading. The letter for the module and ideal is rendered as $\mathcal{J}$; $\mathcal{I}$ or J are possible.

Blow-up — ? M is a manifold; $D \subset M$. [Beneath D:] a subset with singularities.

M_0^H is the blow-up. Blowing-up. The inverse image of D is D', a manifold of dimension $n-1$. D_0 is $(n-1)$-dimensional.



Original source details: GELFAND-P000001

Diagram and typed description. Three enclosed drawings: at left a region with D; at upper right a region with D' and the label M_0^H; at lower right a region and the label M_0'. The upper arrow points left; the lower oblique arrow also points left, with the label “continuous map” [source abbreviation: непр. отобр.]. The precise inner curves, labels and the connection of the lower region to the upper one require comparison with the scan.



Original source details: GELFAND-P000001

Uncertain reading. The upper index on M_0 resembles H; the lower region’s label resembles M_0'.

The construction is Grothendieck’s. A is a ring, $\mathcal{J}$ is a module. Example. A is a ring of functions; $\mathcal{J}$ is the ideal of functions taking the value 0 on D.

One obtains a sheaf of rings and a sheaf of modules. A is a ring, $\mathcal{J}$ is an ideal. Construct X, O_X a sheaf of rings, $\mathcal{J}_X$ a sheaf of ideals, so that

$$(A, \mathcal{J}) \longleftarrow (X, O_X, \mathcal{J}_X).$$

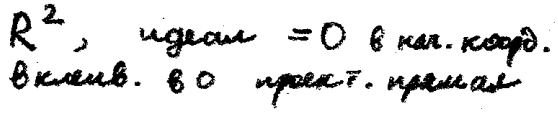
[Beneath the arrow:] linear map.

$(A, \mathcal{J})$ is a sheaf generated by 1 ideal.

$\mathcal{J}_X$ is invertible, i.e. an ideal of the ring: $\exists f \in \mathcal{J}_X$ such that $O_X f = \mathcal{J}_X$. [Above the formula, an arrow to $\exists$:] generator.

(The ideal is principal if $\mathcal{J}$ is an ideal.) Then the zero variety is 1 equation.

A is a ring, $\mathcal{J}$ is an ideal. Typical example: R^2, the ideal $= 0$ at the origin.



Original source details: GELFAND-P000001

Uncertain reading. The next line: “Gluing a projective line into 0” [source: Вклеив. в 0 проект. прямой]; the beginning may read “Оклеив.”.

A, $\mathcal{J}$ are modules.

$$\mathcal{J} \otimes_A \mathcal{J}, \qquad \mathcal{J} \otimes_A \mathcal{J} \otimes_A \mathcal{J}, \dots$$

[Above the groups of tensor products:] symmetric; symmetric.
Consider formal sums

$$\sum_{k=0}^{N} a_k z^k.$$

Add and multiply. $x \otimes y = y \otimes x$.

$$a_0 \in A, \qquad a_1 \in \mathcal{J}, \qquad a_2 \in \mathcal{J} \otimes_A \mathcal{J} \quad —$$

Full reading notes

GELFAND-0001-U001



Original source details: GELFAND-P000001

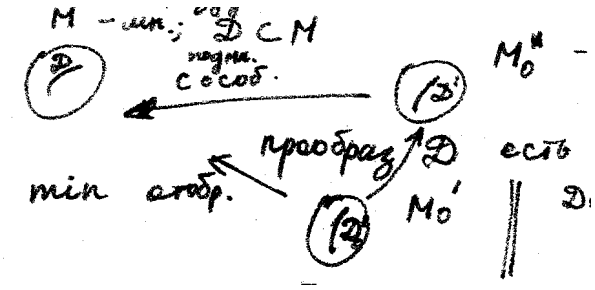
Observation. After “A is a ring” is a handwritten letter with a fork at the top and a lower hook; it recurs for the module and the ideal.

Candidates: $\mathcal{J}$; $\mathcal{I}$; J

Historical note: A calligraphic J was used in the text as a graphical candidate; the notation was not identified from its mathematical meaning.

Fresh source comparison: The same fork and lower hook are visible in three separate occurrences at 5×/10×. This supports a consistent graphical transcription but does not distinguish every possible conventional I/J letterform.

GELFAND-0001-U002



Original source details: GELFAND-P000001

Observation. In the upper half are three rounded contours with inner curves and two leftward arrows; D,D-prime,M and indices occur above and beside them.

Candidates: None proposed.

Historical note: Uncertain inner curves and joins were not redrawn. The description and the source-figure entry retain a locator for comparison.

Fresh source comparison: The complete diagram crop is now displayed at its typed description. Both leftward arrowheads are visible. A separate curved stroke rises from the lower contour toward the area beside D; its exact attachment to the upper contour is not asserted.

GELFAND-0001-U003



Original source details: GELFAND-P000001

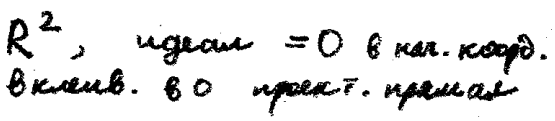
Observation. The upper index on M_0 resembles H; over M_0 at the lower region is a short stroke.

Candidates: M_0^H; M_0'; M_0^N; M_0'

Historical note: The indices and prime need comparison with the original.

Fresh source comparison: Separate close crops show the dark upper index and the lower prime at the reading itself. The lower prime is distinct; the dark upper sign still does not justify excluding N in favor of H.

GELFAND-0001-U006



Observation. To the right of the R^2 example the line begins with a joined group of letters and ends “into 0,a projective line”.

Original source details: GELFAND-P000001

Candidates: Вклеив. в 0 проект. прямой [Gluing a projective line into 0]; Оклеив. в 0 проект. прямой [alternative initial abbreviation]

Historical note: The phrase is given inside the uncertain-reading entry,without choosing from the mathematical meaning of a blow-up.

Fresh source comparison: The wider context and 5×/10× details strongly favor the initial В and the abbreviation Вклеив. The historical alternative remains visible; the source wording is not replaced by a standard construction of a blow-up.